



NIH Cloud Platform Interoperability Administrative Coordinating Center (NCPI-ACC)

2026 Spring Workshop Report

March 4, 2026

Overview

The NIH Cloud Platform Interoperability (NCPI) Program Spring Workshop was held virtually on March 4, 2026. The meeting brought together representatives from the NCPI-funded Interoperability Projects, members of the NCPI Steering Committee, and colleagues from across NIH to discuss lessons learned from NCPI, highlight areas for future work, and foster sustainability. A copy of the agenda can be found in the Appendix.

Sessions included opening remarks from the Office of Data Science and Strategy (ODSS), Administrative Coordinating Center (ACC) highlights, interoperability project presentations, updates on the NCPI Data Catalog and FHIR (Fast Healthcare Interoperability Resources) Working Group, and an “Ask Me Anything” panel with members of the NCPI Steering Committee. Across sessions, a common theme was that NCPI’s progress is real, practical, and powered by people working together on science-driven use cases. Below is a synthesis of the overarching themes, lessons learned, and potential next steps to help translate NCPI’s momentum into durable, scalable practice to advance interoperability followed by more details about each of the presentations and discussions.

Overarching Themes

- People and use cases drive success. The most concrete advances came from projects tightly coupled with specific scientific questions and from teams with sustained cross-platform relationships. Examples of these successes include CAVATICA’s FHIR user interface, the Undiagnosed Disease Network (UDN)–Kids First harmonized analyses, and Watershed’s end-to-end cloud workflows.
- Standards are necessary but not sufficient. Data Repository Services (DRS), Researcher Authentication Services (RAS), and FHIR (and related Global Alliance for Genomics and Health [GA4GH] work) provide the right foundation. However, their utility in practice depends on reference implementations, test suites, and funding to meet non-trivial security and compliance needs.
- Harmonization and metadata are difficult engineering problems. Variable naming, differing pipelines, and diverse consent models require robust transformation tooling and governance. Large language models (LLMs) and artificial intelligence (AI) can accelerate mapping and metadata inference, but only with public metadata, human oversight, and auditable rules.
- Documentation and onboarding must serve multiple audiences. Technical specifications alone will not drive adoption—platform engineers, data stewards, seasoned investigators, and early-career researchers need layered documentation that meets their needs and skill level. Suggestions for materials include one-page overviews, implementation guides, demos, and “citizen-scientist” how-to guides.
- Costs and operational overhead are real and frequently underestimated. Authentication requirements (e.g., RAS and IAL2), privacy/security reviews, data use agreements, egress risks, and ongoing maintenance all require explicit budgeting and transparent cost estimates to inform planning.
- Scaling requires both grassroots pilots and top-down coordination. Science-led pilots create proof points while NIH-level investment and clear operational expectations support scaling across platforms.

Crosscutting Lessons from the Interoperability Projects

- Interoperability patterns that work locally can be generalized: manifests/DRS handoffs (e.g., from AnVIL to Velsera), FHIR viewers and exporters (CAVATICA), and cloudified reproducible workflows (Watershed) are examples of reusable approaches.

- Reciprocal access and DAR workflows can be a bottleneck. The UDN–Kids First work showed that analytic sharing often depends on explicit reciprocal data access requests (DARs), not just platform authentication—this needs templated patterns and clearer legal guidance.
- Workflow portability is essential. Packaging into conda/Docker, publishing workflow description language/common workflow language (WDL/CWL) workflows, and providing cloud workspaces significantly lower the barrier for reuse and reproducibility.
- Small, focused investments can yield outsized results. Playbooks, implementation guides, and a small number of reference implementations enabled multiple teams to move faster than would have been possible by ad hoc effort alone.

Potential Next Steps

Workshop attendees offered a range of next steps that could be undertaken to continue to advance interoperability and support the NCPI ecosystem. These include:

Share Lessons Learned

- Publish an “NCPI Lessons Learned” roadmap summarizing principles, what approaches work, and what do not, including high-level cost and labor estimates for common tasks (security, RAS, FHIR deployment).

Develop Tools and Resources that Meet the Needs of Many Users

- Produce multi-tier documentation for DRS, RAS, and FHIR: one-page summaries, implementation how-tos, and developer references plus short video demos and user stories.
- Deliver runnable reference implementations and test suites for DRS and RAS (and a canonical FHIR IG example) so new adopters have a validated starting point.
- Continue to preserve and steward ACC outputs—playbooks, GitHub assets, dataset catalog entries, public workspaces and cloudified workflows—so other NIH programs can adopt them.

Support Future NIH Planning and Budgeting

- Convene recurring NIH co-innovation events (time-boxed “jamborees”) to surface new scientific use cases and accelerate prototype interoperability across platforms.

- Publish transparent cost and effort templates to guide IC budgeting and reduce surprises during implementation.

Drive Technical Advances that Enable Collaboration

- Advance safe, auditable AI pilots focused on metadata harmonization and variable mapping (public metadata combined with human-in-the-loop), accompanied by playbooks and evaluation criteria.
- Support navigation of access controls and consent requirements to enable workflow portability and support collaboration.
- Standardize reciprocal DAR and derived-product sharing workflows with templates and recommended legal/security language to reduce friction in collaborative analyses.

The tangible outputs and lessons learned demonstrate that interoperable, cross-platform research is achievable and scientifically valuable. The building blocks presented during the workshop will require sustained attention to documentation, reference implementations, predictable funding for compliance and other resource-intensive work, and a combination of bottom-up science pilots and top-down coordination. The NCPI community has already built many of the technical and social building blocks; the next phase is about packaging, sharing, and resourcing those building blocks so more investigators can benefit.

Appendix. NCPI Spring Workshop Agenda

Workshop Objectives

To share successes and lessons learned from the interoperability projects and identify common challenges and potential next steps for the NCPI ecosystem

Agenda

March 4, 2026, 1:00-4:00pm EST

Time	Session	Speakers
1:00-1:05	Welcome & Housekeeping	Alison Banger, NCPI ACC
1:05-1:20	Welcome from ODSS with Q&A	Richard Conroy, NIH ODSS
1:20-1:35	NCPI ACC Highlights	Elizabeth Robbins, NCPI ACC PI
1:35-1:50	Interoperability Project Presentation with Q&A <i>Facilitating understanding of shared disease mechanisms leveraging interoperability standards across dbGaP/SRA, ImmPort, and Kids First DRC</i>	Allison Heath, CHOP
1:50-2:05	Interoperability Project Presentation with Q&A <i>AnVIL-BioData Catalyst interoperability project to leverage GTEx, T2T and HPRC to study rare genetic variants in TOPMed cohorts with deep phenotype data, whole genome sequence data, and RNA-sequence data</i>	Michael Schatz, Johns Hopkins
2:05-2:15	Break	
2:15-2:30	Interoperability Project Presentation with Q&A <i>Integrated data exploration of AnVIL and KidsFirst to enable sex-biased chromosome association and determine the genetic basis of Ollier disease and Maffucci syndrome</i>	Benedict Paten, UCSC
2:30-2:40	NCPI Data Catalog Update	Dave Rogers, Clever Canary
2:40-3:00	Updates from the FHIR and Search Working Groups	Allison Heath, CHOP Robert Carroll, VUMC Jonathan Lawson, Broad Institute
3:00-3:55	Discussion and AMA with the Steering Committee	Valerie Cotton, NCPI SC, Moderator
3:55-4:00	Concluding Remarks	Janey Hsiao, NIH ODSS
4:00	Adjourn	